Group Segregation Caused by Reciprocity Norm

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Abstract

Indirect reciprocity is pointed out as a mechanism to maintain cooperation in society. It requires people to monitor others and find good individuals, whom they cooperate with. Previous studies have mainly focused on whether cooperation evolves with appropriate definition of goodness. Though actual society observes that people are more cooperative inside of their own groups, the structure of cooperative relation (i.e., who cooperates with whom) under indirect reciprocity has not been studied enough. Therefore, we studied a model of indirect reciprocity with private assessment (i.e., people can have different opinions on who are good) and identified what structure emerges. It is revealed that the specific types of reciprocity norm cause split of society into multiple groups only inside of which mutual cooperation is sustained. This result suggests reciprocity, which usually promotes cooperation, can cause segregation of society.

Keyword: cooperation, reciprocity, group formation

Mutual cooperation is a foundation of human society. Mechanisms that sustain cooperative behavior have been studied in many fields of social science. Indirect reciprocity is pointed out as one of the mechanisms. It requires people to monitor the interaction of others and find good individuals, whom they cooperate with. Previous studies have mainly focused on what definition of goodness promotes evolution of cooperation[1, 2].

In contrast with evolution of cooperation itself, the structure of cooperation (i.e., who cooperates with whom) has not been paid much attention to in studies of reciprocity norms. However, actual society observes very heterogeneous structure of cooperation. For example, people are more cooperative inside of their own groups, which is called in-group favoritism.

Some recent studies pointed out the connection between in-group favoritism and indirect reciprocity (e.g. [3]). These studies focused the case that group structure itself already exists and therefore are applicable to in-group favoritism based on, for example, races, ethnicities, and nationalities, which people cannot change easily.

However, in our society, there are various situations in which group structure is not determined a priori. For example, it is not likely that groups of friends in school are completely determined before they meet first in the beginning of school. Structure of cooperation in such context is established through their interaction and in-group favoritism emerges even in such case. Therefore, we need a model which explains both of group formation and evolution of cooperation.

Our hypothesis was that even without exogenous group structure, indirect cause in-group favoritism with endogenous group structure. Therefore, we investigated the structure of cooperation of people who follow the reciprocity norms without assuming any group structure in the initial state. Our study revealed that the specific types of reciprocity norm cause spontaneous split of society into multiple groups only inside of which mutual cooperation is sustained [4]. This result suggests reciprocity, which usually promotes cooperation, can cause segregation of society.

References

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